

STARLight × NVIDIA

Objective

A media house has asked for the founder's perspective on the STARLight–NVIDIA development: its market impact, whether it's a genuine breakthrough, the hardware stakes, and what it signals for the industry.

Get this right first

STARLight is not an NVIDIA–STMicroelectronics joint venture. It's an EU CHIPS-funded consortium of 24 companies and universities across 11 countries, led by STMicroelectronics, running through 2028. NVIDIA is one contributing partner, not a co-founder. Leading with this framing helps avoid a mischaracterisation.

Research & business analysis

Impact on shares

NVDA fell 2.82% the day STARLight was announced (23 Sept 2025) and a further 0.82% the next day. That's the opposite of what a "this is bullish for NVIDIA" narrative would predict, and it suggests markets read it as a multi-year infrastructure story rather than a catalyst. STM's exposure is more direct: it entered high-volume PIC100 production in March 2026 with plans to quadruple capacity by 2027, and has re-rated far more over the past year than NVDA. At NVIDIA's ~\$5T scale, even a meaningful photonics tailwind is immaterial to the multiple; at STM's ~\$46B scale, it's structural.

Groundbreaking, or industry trend?

Industry trend, not invention. Silicon photonics has been the acknowledged next step in AI networking since NVIDIA's own March 2025 unveiling of Spectrum-X and Quantum-X. What STARLight adds is the industrialization layer: 300mm wafer production, packaging, and a sovereign EU supply chain, not new physics.

Hardware significance

AI scaling has become an interconnect problem as much as a compute one. Co-packaged optics cut electrical signal loss from roughly 22dB to 4dB and claim 3.5x power efficiency over pluggable transceivers. GPUs are only as effective as the fabric connecting them, and that fabric is now the harder thing to manufacture at scale.

3.5x

power efficiency vs. pluggable optics

22→4dB

electrical signal loss, CPO vs. traditional

24 / 11

partners / EU countries in STARLight

4x

ST's planned PIC100 capacity by 2027

Talking points & sources

Talking points

- 1 Interconnect speed is becoming the new AI moat. The race is shifting from a pure chip-performance contest to an infrastructure bottleneck problem. A faster GPU means little if the fabric linking thousands of them is choked. As clusters scale toward millions of GPUs, moving data between them efficiently matters as much as the processors themselves.
- 2 Silicon photonics is reaching commercial maturity, not a new breakthrough. It's been called "a few years away" for the better part of a decade. What's changing isn't the physics (that's been settled since NVIDIA's own Spectrum-X/Quantum-X unveiling in March 2025); it's the manufacturing. STARLight is the 300mm, high-volume European supply chain that turns co-packaged optics into a shipped product.
- 3 It's solving a power problem as much as a speed problem. Data-center power draw is pushing grids toward their limits. Moving from electrical to optical signaling inside the fabric, which is roughly 3.5x more power-efficient than pluggable transceivers, is one of the effective efficiency gains available to keep scaling AI clusters.
- 4 NVIDIA's competitive footprint is widening, and no one owns this stack alone. The value proposition now spans GPUs, networking, and optics, not compute alone. NVIDIA's role as one of 24 STARLight partners shows a dominant US technology player leaning on European semiconductor manufacturing to build the next layer of AI infrastructure. That's interdependence, not a bilateral deal.
- 5 Commercialization is the number that matters. ST moving PIC100 to high-volume production, with capacity plans to quadruple by 2027, is a stronger signal than any single partnership announcement. It's evidence that the industrial pipeline is moving beyond the lab toward production scale.

Proposed founder quote

"Silicon photonics isn't a new direction for our industry. That's been clear for years. What's changing is scale and economics. As AI clusters move toward millions of GPUs, the network connecting them becomes as critical as the processors themselves, and the power it draws becomes just as critical as its speed. STARLight matters because it's an EU-led effort, and NVIDIA is one of twenty-four partners in it, focused on turning co-packaged optics into high-volume manufacturing rather than a lab demo. The real story isn't a partnership. It's optical interconnects becoming a mainstream, industrialized part of AI infrastructure."

Sources

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